AMBITION GURU

Computer Graphics (CSIT 401)

IV semester, BScCSIT

Compiled by Ankit Bhattarai
Ambition Guru College

Syllabus

Unit	Contents	Hours	Remarks
1.	Introduction to Computer Graphics	5	
2.	Scan Conversion Algorithms	7	*Very Important*
3.	2D Transformations and Viewing	6	*Very Important*
4.	**3D Transformations and Viewing**	**5**	
5.	3D Object Representation	6	
6.	Solid Modeling	3	
7.	Visible Surface Detection	5	
8.	Illumination, Shading and Rendering Techniques	5	
9.	OpenGL and Graphics Programming	3	

Practical Works

Credit hours : 3

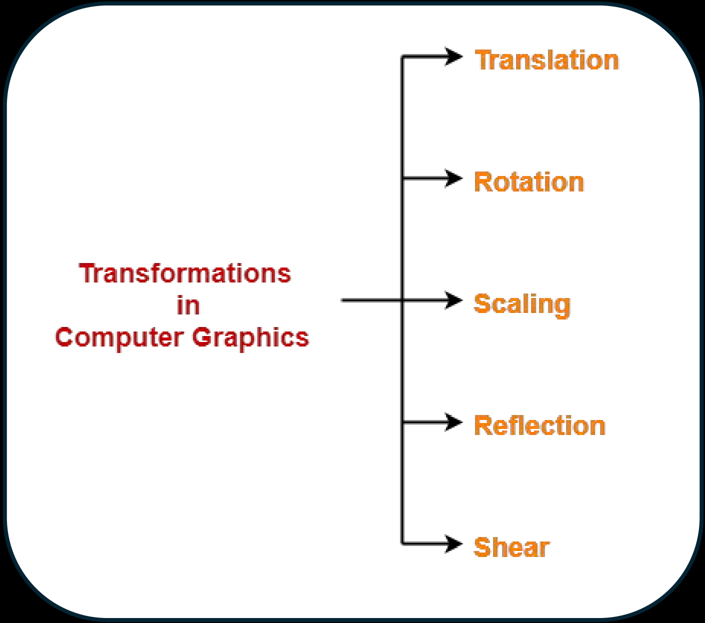
Unit 4
3D Transformations and Viewing
(5 hrs.)

3D co-ordinate systems, 3D transformations: Translation, rotation, scaling, affine transformations, composite transformation, 3D viewing pipeline: world, viewing, and screen coordinates, Projection techniques: orthographic, parallel, perspective projections.

3D Geometric Transformation

3D Transformation

- When the transformation *takes place on a 3D plane*. It is called 3D transformation.
- Generalize from 2D by including **z** coordinate
- Straight forward for translation and scale, rotation more difficult



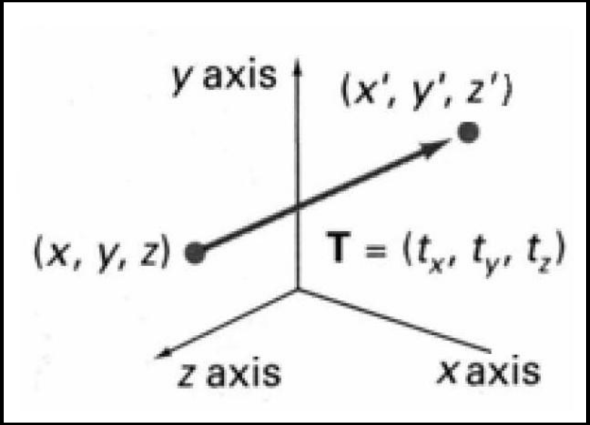
Homogeneous coordinates: 4 components

Transformation matrices: 4×4 elements

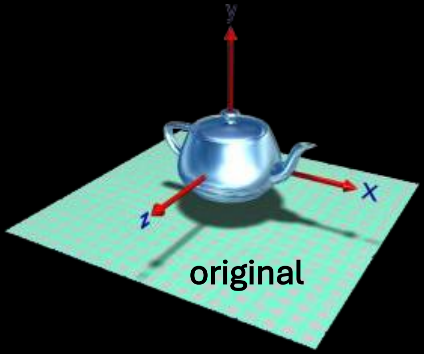
3D Translation

- Moving of object is called translation.
- In 3 dimensional homogeneous coordinate representation , a point is transformed from position P = (x, y , z) to P'=(x', y', z')
  - This can be written as: Using P'= T . P where

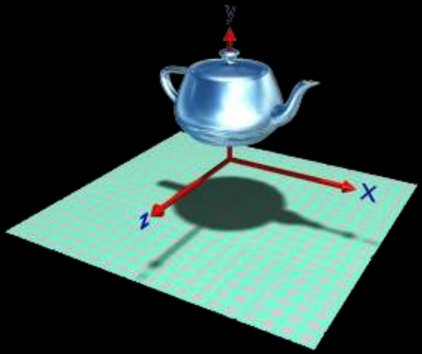
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Translation



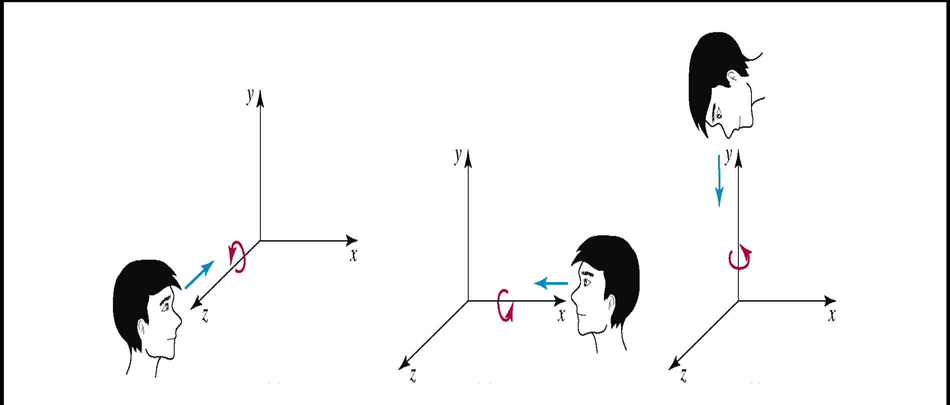
original



translation along y, or
V = (0, k, 0)

3D Rotation

- Where an object is to be rotated about an axis that is parallel to one of the coordinate axis, we can obtain the desired rotation with the following transformation sequence.



Coordinate axis rotation

✓ Z- axis Rotation
✓ Y-axis Rotation
✓ X-axis Rotation

3D Rotation

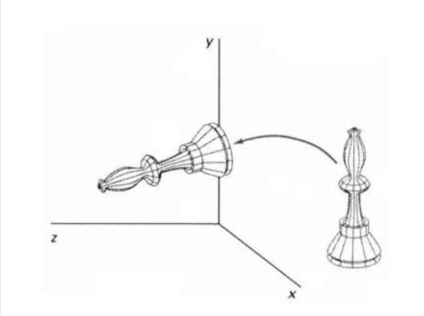
AMBITION GURU

The equation for X-axis rotation

x' = x

y' = y cosθ – z sinθ

z' = y sinθ + z cosθ



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Compiled by ab

3D Rotation

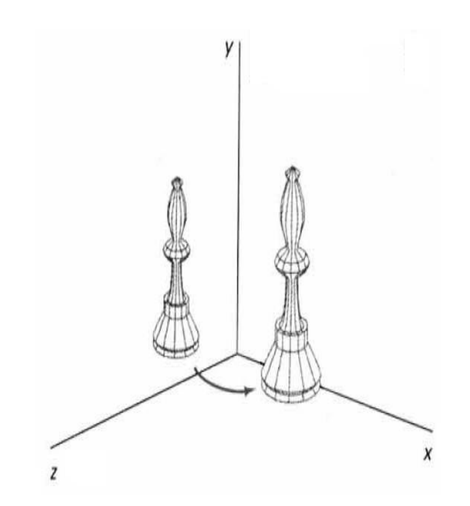
AMBITION GURU

The equation for Y-axis rotaion

x' = x cosθ + z sinθ

y' = y

z' = z cosθ - x sinθ



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Rotation

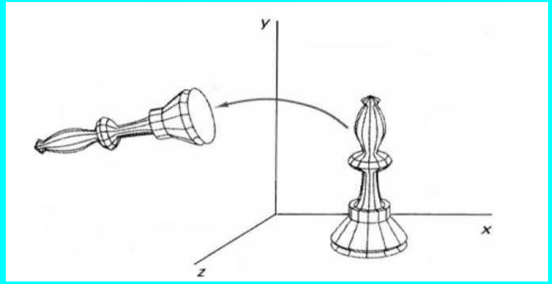
The equation for Z-axis rotation

$x' = x \cos\theta - y \sin\theta$

$y' = x \sin\theta + y \cos\theta$

$z' = z$

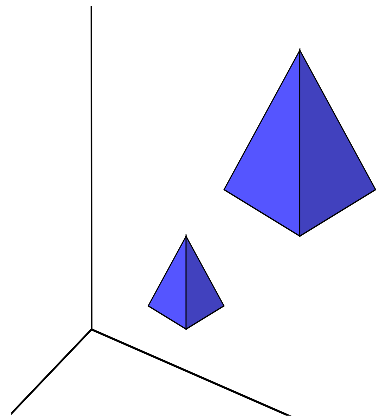
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Scaling

- Changes the size of the object and repositions the object relative to the coordinate origin.
- In the scaling process , you either expand or compress the dimensions of the object.
- Scaling can be achieved by multiplying the original coordinates of the object with scaling factor to get the desired result

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

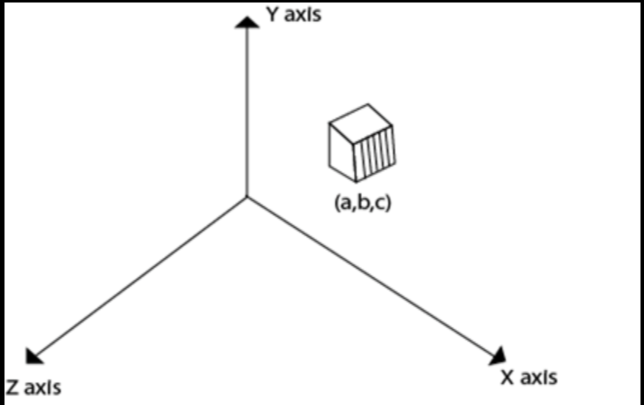


3D Scaling: Scaling with respect to a selected fixed position

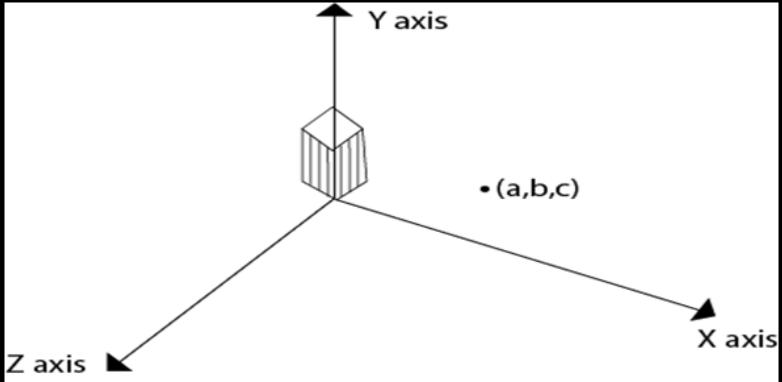
Following are steps performed when scaling of objects with fixed point (a, b, c). It can be represented as below:

1. Translate fixed point to the origin
2. Scale the object relative to the origin
3. Translate object back to its original position.

In figure (a) point (a, b, c) is shown, and object whose scaling is to done also shown in steps in fig (b), fig (c) and fig (d).

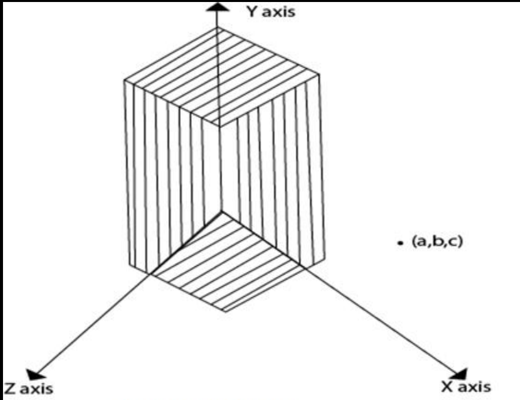


Object & point for scaling
(a)

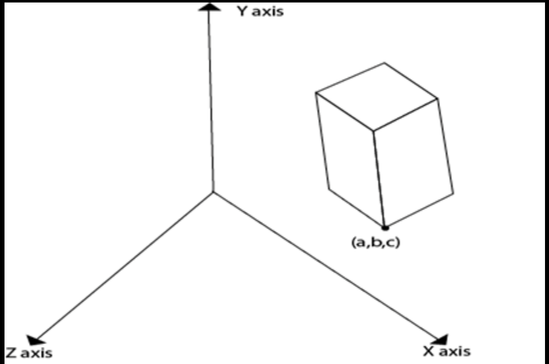


Object transfer at origin
(b)

3D Scaling: Scaling with respect to a selected fixed position



Scaling of object
(c)



Object shifted to original position after scaling
(d)

The matrix representation for an arbitrary fixed-point scaling can then be expressed as the concatenation of these translate-scale-translate transformations as:

$$\mathbf{T}(x_f, y_f, z_f) \cdot \mathbf{S}(s_x, s_y, s_z) \cdot \mathbf{T}(-x_f, -y_f, -z_f) = \begin{bmatrix} s_x & 0 & 0 & (1 - s_x)x_f \\ 0 & s_y & 0 & (1 - s_y)y_f \\ 0 & 0 & s_z & (1 - s_z)z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Reflection

- Reflection in computer graphics is used to emulate reflective objects like mirrors and shiny surfaces
- Reflection may be an x-axis y-axis , z-axis. and also in the planes xy-plane,yz-plane , and zx- plane.

Note: Reflection relative to a given Axis are equivalent to 180 Degree rotations.

3D Reflection

AMBITION GURU

- Reflection about xy-plane(or z=0plane)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Reflection about yz-plane(or x=0plane)

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Reflection

AMBITION GURU

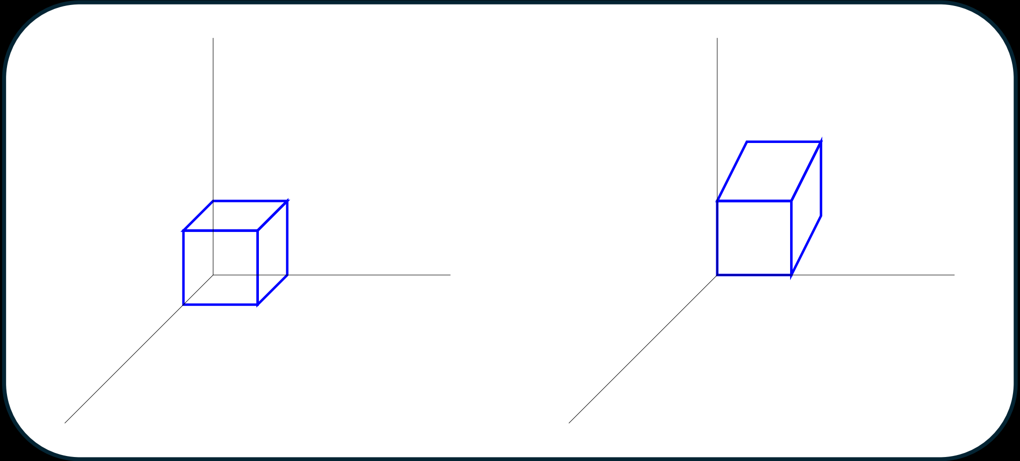
- Reflection about zx-plane(or y=0plane)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Shearing

- Modify object shapes
- Useful for perspective projections
- When an object is viewed from different directions and at different distances, the appearance of the object will be different.
- Such view is called perspective view.
- Perspective projections mimic what the human eyes see.

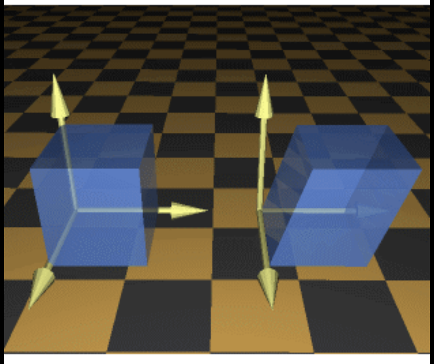
e.g. draw a cube (3D) on a screen (2D) Alter the values for **x** and **y** by an amount proportional to the distance from z_{ref}



3D Shearing

- Matrix for 3d shearing
- Where a and b can be assigned any real value.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



- In (y, z) w.r.t. x value $SH_{yz} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ sh_y & 1 & 0 & 0 \\ sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- In (z, x) w.r.t. y value $SH_{xz} = \begin{bmatrix} 1 & sh_x & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- In (x, y) w.r.t. z value $SH_{xy} = \begin{bmatrix} 1 & 0 & sh_x & 0 \\ 0 & 1 & sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Practice Question

Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

- Old corner coordinates of the triangle = A (0, 0, 0), B(1, 1, 2), C(1, 1, 3)
- Shearing parameter towards X direction (Sh_x) = 2
- Shearing parameter towards Y direction (Sh_y) = 2
- Shearing parameter towards Z direction (Sh_z) = 3

P' = Shear * P

4 x 4 matrix→ Shear
4 x 1 matrix → Coordinate

Shearing in X Axis-

New coordinates of corner A after shearing = (0, 0, 0)

New coordinates of corner B after shearing = (1, 3, 5)

New coordinates of corner C after shearing = (1, 3, 6)

New coordinates of the triangle after shearing in X axis = A (0, 0, 0), B(1, 3, 5), C(1, 3, 6)

Hint:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ sh_y & 1 & 0 & 0 \\ sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Practice Question

Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Shearing in Y Axis-

New coordinates of corner A after shearing = (0, 0, 0)

New coordinates of corner B after shearing = (3, 1, 5)

New coordinates of corner C after shearing = (3, 1, 6)

New coordinates of the triangle after shearing in Y axis = A (0, 0, 0), B(3, 1, 5), C(3, 1, 6)

Practice Question

Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Shearing in Z Axis-

New coordinates of corner A after shearing = (0, 0, 0)

New coordinates of corner B after shearing = (5, 5, 2)

New coordinates of corner C after shearing = (7, 7, 3)

New coordinates of the triangle after shearing in Y axis = A (0, 0, 0), B(5, 5, 2), C(7, 7, 3)

Question:

Given a homogeneous point (1, 2, 3). Apply rotation 90 degree towards X, Y and Z axis and find out the new coordinate points.

Projection

Introduction to Projection

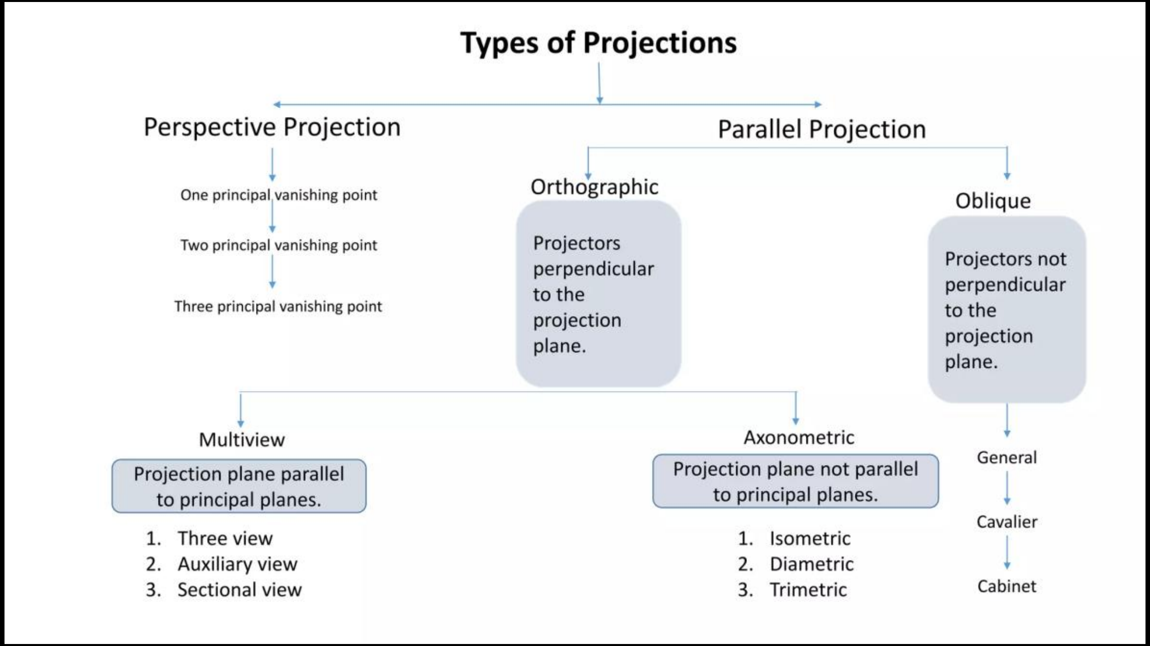
- Projection means transformation that changes a point in n-dimensional coordinate system into a point in a coordinate system that has dimension less than n. In other words, Projection are defined as mapping of three-dimensional points to a two-dimensional plane.
- Converts 3-D viewing co-ordinates to 2-D projection co-ordinates.
- View Plane or Projection Plane: Two-dimensional plane in which 3D objects are projected is called the view plane or projection plane. Simply it is a display plane on an output device.

Introduction to Projection

- **Center of projection**: Point from where projection is taken.
- **Projection plane**: the plane on which the projection of the object is formed.
- **Projectors**: Lines emerging from the center of projection and hitting the projection plane
- **Parallel projection**: In this projection, the co-ordinate positions are transformed to the view plane along parallel lines.
- **Perspective Projection**: In this projection, the co-ordinate positions are transformed to the view plane along lines that converges to a point called as center of projection.

Introduction to Projection

AMBITION GURU



Parallel Projection

- Orthographic parallel projection
- Oblique parallel projection

Perspective Projection

Introduction to Projection : Parallel Projection

AMBITION GURU

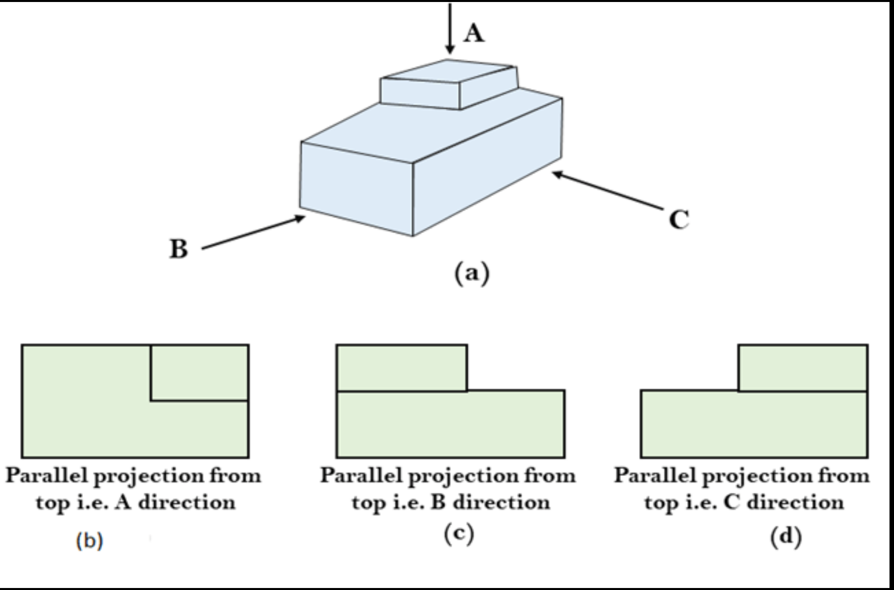
- Coordinate positions are transformed to view plane along parallel lines (projection lines)
- Preserves relative proportions of objects
- Accurate views of various sides of an object are obtained.
- Doesn’t give realistic representation of the appearance of the 3-D object

Types

✓ **Orthographic**- when the projection is perpendicular to the view plane. Used to produce Front, Side and Top view of an object

✓ **Oblique** – when the projection is not perpendicular to the view plane

Introduction to Projection : Parallel Projection



Parallel Projection : Orthographic

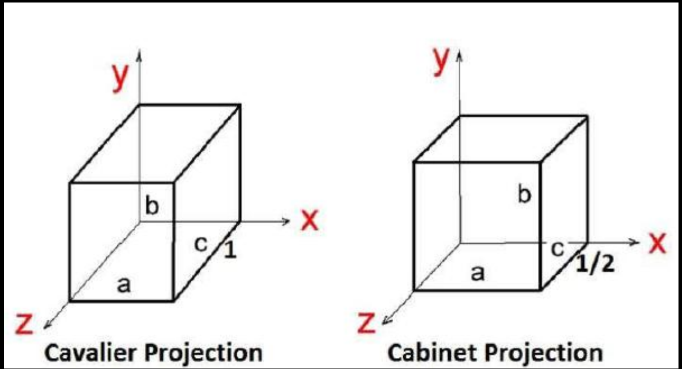
Introduction to Projection : Oblique Projection

AMBITION GURU

- A projection in which the angle between the projectors and the plane of plane of projection is not equal to 90^0 is called oblique projection.
- An oblique projection is formed by parallel projections from a center of projection at infinity that intersects the plane of projection at an oblique angle.

✓ **Cavalier:** All lines perpendicular to the projection plane are projected with no change in length.

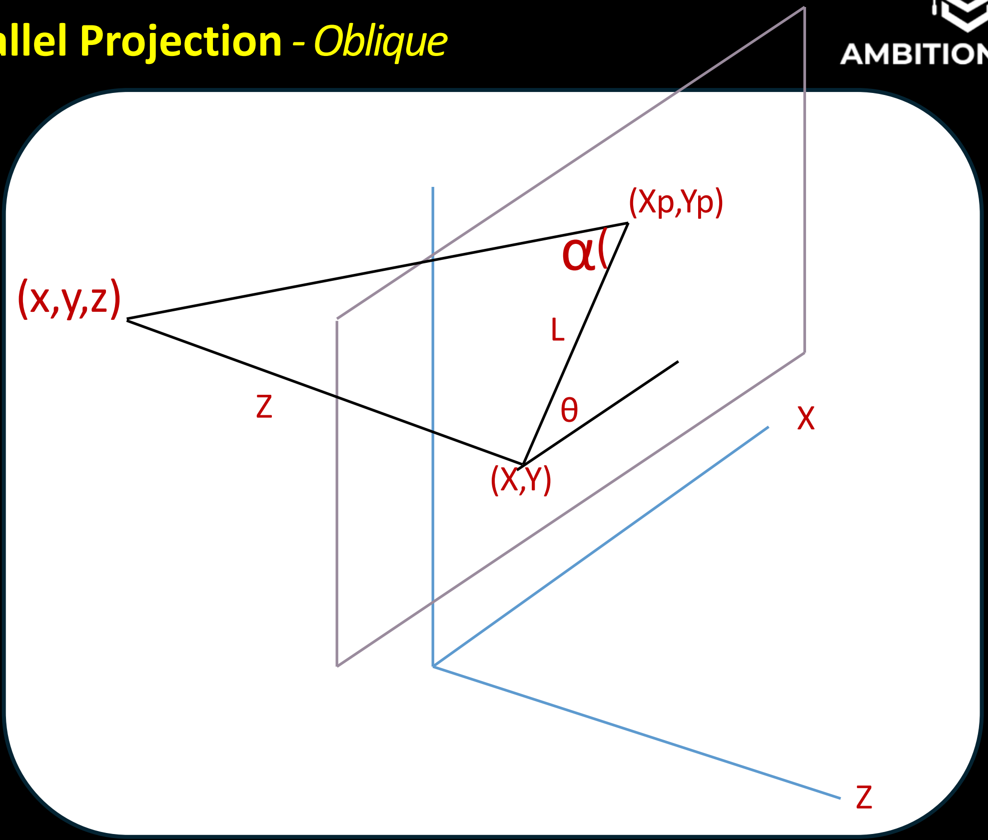
✓ **Cabinet:** All lines perpendicular to the projection plane are projected to one half of their length. These give a realistic appearance of object.



Introduction to Projection: Parallel Projection - *Oblique*

AMBITION GURU

- Point (x, y, z) is projected to position (x_p, y_p) on the view plane.
- Orthographic projection coordinated on the plane are (x, y). Oblique projection line from (x, y, z) to (x_p , y_p) makes an angle with the line on the projection plane that joins (x_p , y_p) and (x , y).
- This line of length L is at an angle θ with the horizontal direction in the projection plane



Introduction to Projection: Parallel Projection - *Oblique*

- Not perpendicular view. (x,y,z) is projected To position (Xp,Yp) on the view plane.

Cos θ = Xp / L Xp = L Cos θ

But exact position is

Xp = X + L Cos θ

Similarly,

Sin θ = Yp / L

Yp= Y + L Sin θ

L depend on angle **α**

Tan α = Z / L

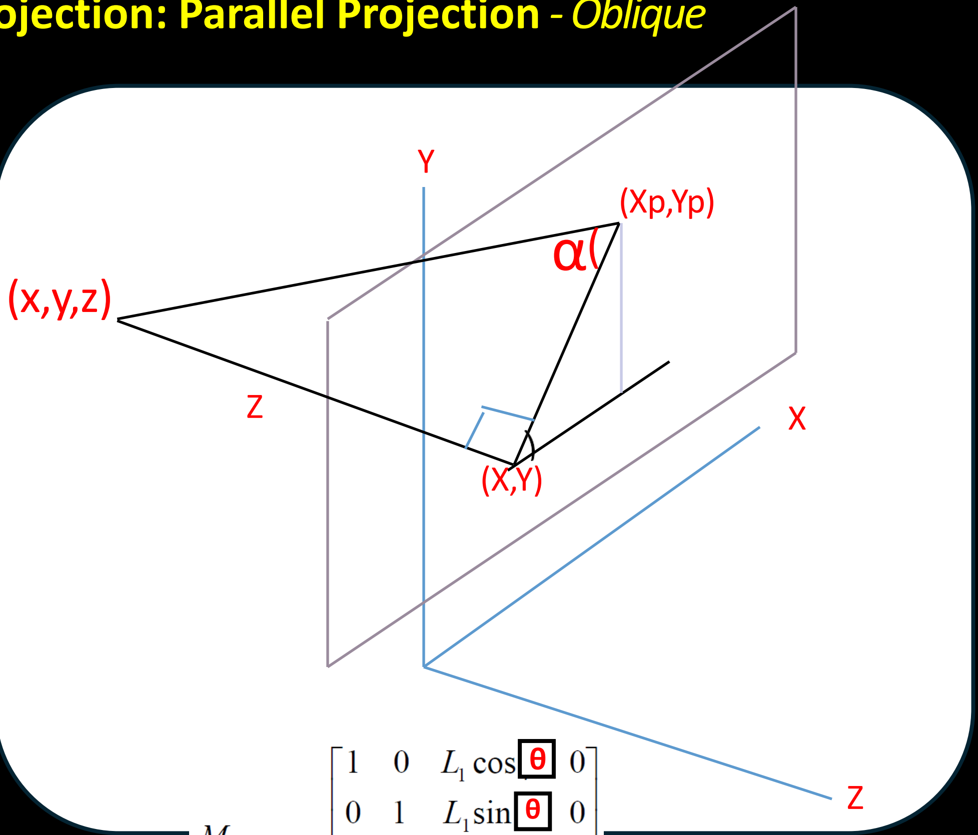
L= Z L_1

Where L_1 is inverse of tan α

Xp = X +Z L_1 Cos θ

Yp= Y + Z L_1 Sin θ

When α = 0 , i.e. L_1= 0 , it is orthographic projection



$$M_{parallel} = \begin{bmatrix} 1 & 0 & L_1 \cos \boxed{\theta} & 0 \\ 0 & 1 & L_1 \sin \boxed{\theta} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Introduction to Projection: Perspective Projection

- In this projection, the co-ordinate positions are transformed to the view plane along lines that converges to a point called as center of projection.
- Two main characteristics of perspective are **vanishing points** and **perspective foreshortening**.
- Due to foreshortening object and lengths appear smaller from the center of projection. More we increase the distance from the center of projection, smaller will be the object appear.
- The projection is perspective **if the distance is finite or has fixed vanishing point or the incoming rays converge at a point**. Thus, the viewing dimension of object is not exact i.e. seems large or small according as the movement of the view plane from the object.

Introduction to Projection: Perspective Projection

- **One Point**: There is only one vanishing point as shown in fig (a)
- **Two Points**: There are two vanishing points. One is the x-direction and other in the y -direction as shown in fig (b)
- **Three Points**: There are three vanishing points as in fig (c).
- In Perspective projection lines of projection do not remain parallel. The lines converge at a single point called a center of projection. The projected image on the screen is obtained by points of intersection of converging lines with the plane of the screen.
- The image on the screen is seen as of viewer's eye were located at the center of projection, lines of projection would correspond to path travel by light beam originating from object.

Introduction to Projection: Perspective Projection

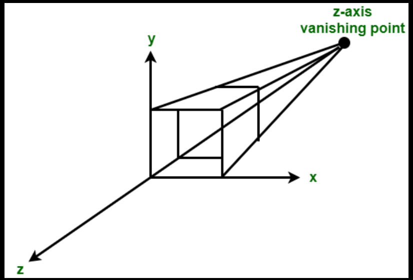


Fig (a) One point

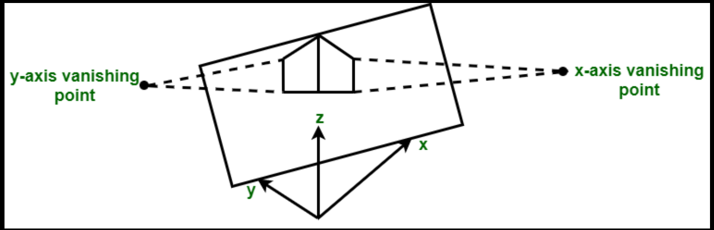


Fig (b) Two point

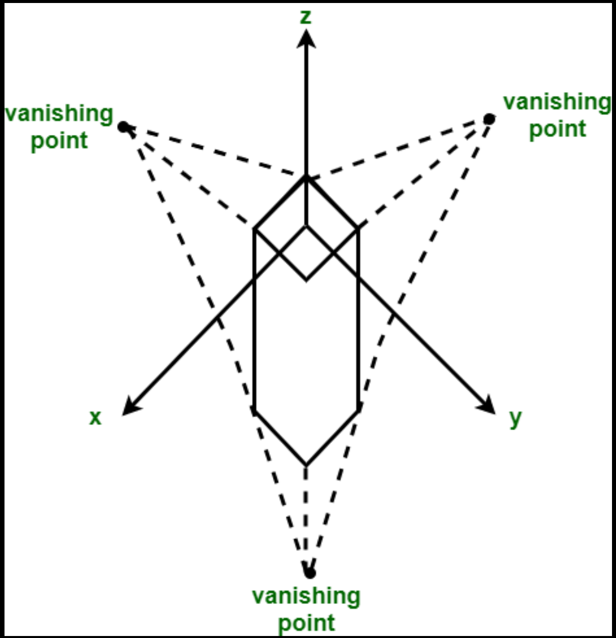
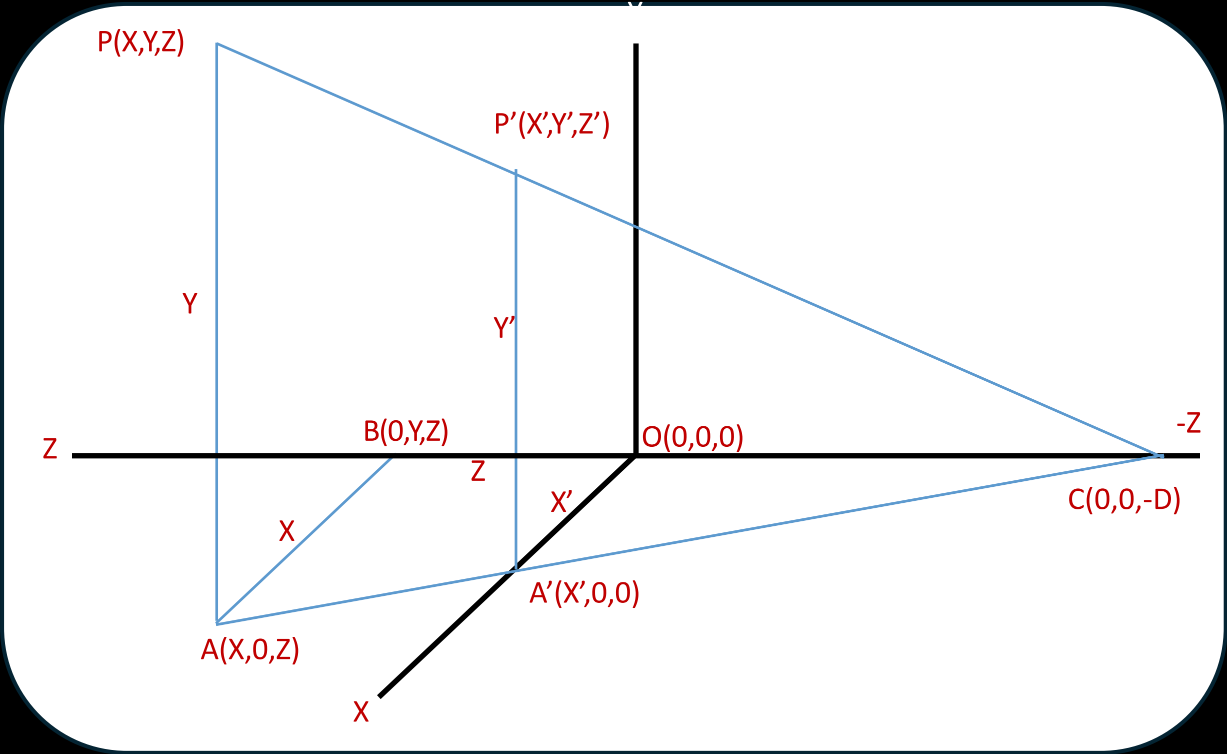


Fig (c) Three point

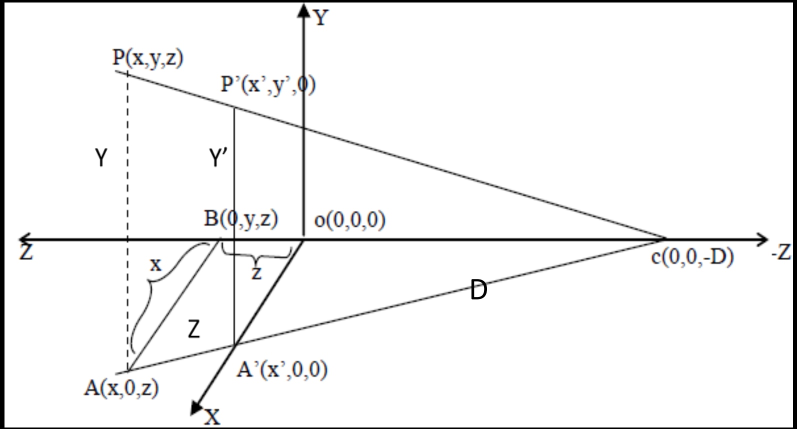
Perspective



Here, center of Projection is c (0,0,-D) along the direction of Z-axis, so the reference point is taken of world coordinate space Wc and the normal vector N is aligned with the y axis.

So, now the view plane vp is the xy plane and center of projection is c (0,0,-D) now from similar triangles ABC and A'OC.

Perspective



Triangles ABC and A'OC
(x/x') = AC/A'C = (Z+D)/D
X' = (XD)/(Z+D)
And Z'=0

Triangles APC and A'P'C
(y/y') = (AC/AC') = (Z+D)/D
y' = (DY)/(Z+D)
And Z' =0

Now, in homogenous coordinates:

$$\begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix} = \frac{1}{z+D} \begin{bmatrix} Dx \\ Dy \\ 0 \\ z+D \end{bmatrix} = \frac{1}{z+D} \begin{pmatrix} D & 0 & 0 & 0 \\ 0 & D & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & D \end{pmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Introduction to Projection

	Parallel Projection	Perspective Projection
Representation of Objects	Parallel projection represents the object in a different way like telescope.	It represents any given object in a three-dimensional manner.
Shape and Size of Objects	It does not alter the shape or the size of the given object on a plane.	In this perspective, the objects that stay far away appear to be smaller in size, while the ones near to the viewer's eyes appear bigger in size.
Distance from Center of Projection	The distance of the given object is infinite from the center of the projection	The distance of the given object is finite from the center of the projection.
Accuracy of View	It can provide a user with an accurate view of the given object.	It cannot provide a user with an accurate view of the given object. The shapes and sizes of the projection tend to differ from its origination.

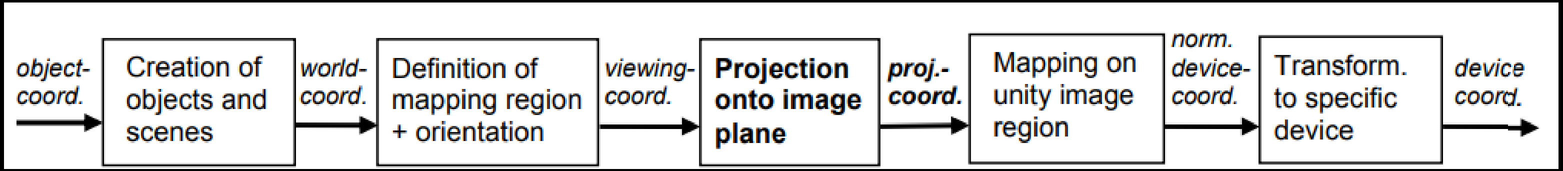
Introduction to Projection

AMBITION GURU

	Parallel Projection	Perspective Projection
Lines of Projection	The parallel projection lines are parallel to each other.	The perspective projection lines are not parallel to each other.
Projector	The projector is also parallel	The projector is not at all parallel.
Types of Projection	There are basically two types of parallel projections: Oblique & Orthographic	There are basically three types of perspective projections : One Point, Two Point & Three Point
Realistic View	Parallel Projection does not form a realistic view of the world and its objects.	Perspective Projection generates a very realistic view of the world and the objects present in it.

3D-Viewing Pipeline

- The viewing-pipeline in 3 dimensions is almost the same as the 2D viewing pipeline. Only after the definition of the viewing direction and orientation (i.e., of the camera) an additional projection step is done, which is the reduction of 3D-data onto a projection plane:



Modeling Transformation → Viewing Transformation → Projection Transformation → Normalization Transformation And Clipping → Viewport Transformation

3D-Viewing Pipeline

- Once the scene has been model, world coordinates position is converted to viewing coordinates.
- The ***viewing coordinates system*** is used in graphics packages as a reference for specifying the observer viewing position and the position of the projection plane.
- ***Projection operations*** are performed to convert the viewing coordinate description of the scene to coordinate positions on the projection plane, which will then be mapped to the output device.
- Objects outside the viewing limits are clipped from further consideration, and the remaining objects are processed through visible surface identification and surface rendering procedures to produce the display within the device viewport.

Questions

AMBITION GURU

1. Given a 3D object with coordinate points A(0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0). Apply the translation with the distance 1 towards X axis, 1 towards Y axis and 2 towards Z axis and obtain the new coordinates of the object. ***Answer:*** A = (1, 4, 3), B = (4, 4, 4), C = (4, 1, 2)
2. Given a homogeneous point (1, 2, 3). Apply rotation 90 degree towards X, Y and Z axis and find out the new coordinate points. ***Answer:*** X-axis rotation: (1, -3, 2), Y-axis rotation: (3, 2, -1), Z-axis rotation: (-2, 1, 3)
3. Given a 3D object with coordinate points A(0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0). Apply the scaling parameter 2 towards X axis, 3 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object. **Answer:** A(0, 9, 9), B(6, 9, 18), C(6, 0, 3), D(0, 0, 0)
4. Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XZ plane and find out the new coordinates of the object. ***Answer:*** New coordinates of the triangle after reflection = A (3, -4, 1), B(6, -4, 2), C(5, -6, 3).
5. Define projection. Differences between parallel and perspective projection.
6. Explain 3D viewing pipeline in brief.
7. Differentiate between parallel and perspective projection with suitable diagram. Illustrate the window to viewport transformation with example.

AMBITION GURU

THANK YOU

Any Queries ?